

6/3 Group Meeting 中橋 一 朗 Hajime Nakahashi

11

Abelian varieties and theta functions

ref: [M] Mumford

[B-L] Birkenhake - Lange

[P] Polishchuk.

+ Mumford { "Tate Lectures on Theta I"
"On the equations Defining abelian Varieties I"

Recall (last time)

An abelian var is projective (from view point of alg var?)

→ Q. How do we get equations defining Abelian var?

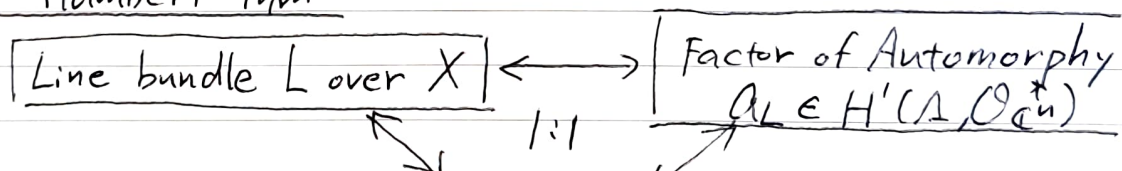
⇒ We need "Theta function" = holomorphic section of an abelian var

Today via analytic theory!

we investigate complex tori $X = \mathbb{C}^n / \Lambda$ where Λ : lattice in $\mathbb{C}^n$

§1 Apell-Humbert thm

Idea



Apell-Humbert datum
(H, χ) where H is Hermitian form
 χ is semi character

§1-1 Factor of Automorphy

Thm $H^p(\mathbb{C}^n, \mathcal{O}_{\mathbb{C}^n}) = 0 \quad p > 0$

pf By $\bar{\partial}$ -Poincaré lemma (see Griffiths-Harris p.46) #

Now we consider s.e.s $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_{\mathbb{C}^n} \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}_{\mathbb{C}^n}^* \rightarrow 1$

and we also have

Thm $H^p(\mathbb{C}^n, \mathbb{Z}) = 0 \quad p > 0$

then we have

Thm $H^p(\mathbb{C}^n, \mathcal{O}_{\mathbb{C}^n}^*) = 0 \quad p > 0$

In particular all holomorphic line bundles on $\mathbb{C}^n$ are trivial.

Let L be a line bundle on X , we choose α

$$\alpha: \pi^*(L) \xrightarrow{\sim} \mathbb{C}^n \times \mathbb{C}$$

Since $\Lambda \cong \mathbb{C}^n$

$$\begin{aligned} \xrightarrow{\sim} \Lambda \cong \pi^* L & \text{ canonical action (i.e. } \pi^*(L)/\Lambda = L) \\ \xrightarrow{\sim} \Lambda \cong \mathbb{C}^n \times \mathbb{C} \end{aligned}$$

(Since the only holomorphic automorphism of a line bundle fixing the base are given by the multiplication by non vanishing holomorphic functions.)

We can write $\phi_\lambda(\lambda, \mu, t) = (\lambda + \mu, f(\lambda, \mu) t)$ where $f(\lambda, \mu) \in \mathcal{O}_{\mathbb{C}^n}^*$

f satisfies "1-cocycle relation"

$$f(\lambda + \mu, \nu) = f(\lambda, \mu + \nu) f(\mu, \nu)$$

$$\begin{array}{ccc} & & \mu + \lambda + \nu \\ & \nearrow & \uparrow \\ \lambda + \mu & & \lambda \\ \downarrow & \xrightarrow{\mu} & \downarrow \\ \nu & & \nu + \mu \end{array}$$

$Z^1(\Lambda, \mathcal{O}_{\mathbb{C}^n}^*) := \{ \text{group of cocycle} \}$ w/ multiplication.
 $\downarrow$
 "factor of Automorphy"

Further, factors of form $(\lambda, \mu) \mapsto \frac{h(\lambda + \mu)}{h(\lambda)h(\mu)}$ for some $h \in \mathcal{O}_{\mathbb{C}^n}^*$ are called "boundary" ($\Leftrightarrow$ this corresponding to $\alpha \mapsto h\alpha$)

Then we can define $H^1(\Lambda, \mathcal{O}_{\mathbb{C}^n}^*) := Z^1/B^1$.

Conversely If we start with a 1cocycle f
 $\Rightarrow$ we can define L on X as $\mathbb{C}^n \times \mathbb{C} / \Lambda$

$$(\lambda, t) \mapsto (\lambda + \mu, f(\lambda, \mu) t)$$

Thus we have

$$\boxed{\phi: H^1(\Lambda, \mathcal{O}_{\mathbb{C}^n}^*) \xrightarrow{\sim} H^1(X, \mathcal{O}_X^*) = \text{Pic}(X)}$$

§1-2 Coboundary map.

$$\text{Since } 0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 1$$

$$\text{we have } \boxed{c_1: H^1(X, \mathcal{O}_X^*) \rightarrow H^2(X, \mathbb{Z}) \text{ first chern class}}$$

On the other hand, similarly we have

$$\boxed{\delta: H^1(\Lambda, \mathcal{O}_{\mathbb{C}^n}^*) \rightarrow H^2(\Lambda, \mathbb{Z})}$$

Abut group cohomology
see Akita Toshiyuki's
surveys.

By the same discussion in §1-1 we also have

$$\text{Thm } \phi: H^i(\Lambda, \mathbb{Z}) \xrightarrow{\sim} H^i(X, \mathbb{Z}) \text{ for all } i > 0.$$

Pf See [M] Appendix to §2 or [BL] Appedix B. #

Then we get

$$H^1(X, \mathcal{O}_X^*) \xrightarrow{\sim} H^1(\Lambda, \mathcal{O}_{\mathbb{C}^n}^*) \ni f_\lambda = \exp(2\pi i g_\lambda)$$

$$c_1 \downarrow$$

$$\downarrow \delta$$

$$\downarrow$$

$$H^2(X, \mathbb{Z}) \xrightarrow{\sim} H^2(\Lambda, \mathbb{Z}) \quad \delta f = t_\lambda^* g_\mu - g_{\lambda+\mu} + g_\lambda$$

$$* \downarrow \delta$$

$$\Lambda^2 \text{Hom}(\Lambda, \mathbb{Z})$$

well defined by 1-cocycle rels

* we use $F(\lambda, \mu) \mapsto F(\lambda, \mu) - F(\mu, \lambda)$

then $\boxed{E_L := t_\lambda^* g_\mu - t_\mu^* g_\lambda + g_\lambda - g_\mu}$ is skew symmetric form is determined for each line bundle L on X .

$$E_L: \Lambda \times \Lambda \rightarrow \mathbb{Z} \xrightarrow{\mathbb{R}\text{-lin. ext}} E: \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{R}$$

Pwp ([BL] Pwp 2.16)

$E: \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{R}$ skew sym form. TFAE

(1) $\exists L \in \text{Pic}(X)$ s.t. E represents $C_1(L)$

(2) $E(\Delta, \Delta) \leq \mathbb{Z}$ & $E(iu, iw) = E(u, w)$ for all $u, w \in \mathbb{C}^n$

pf we consider comm. diag.

$$H^1(\mathcal{O}_X^*) \xrightarrow{C_1} H^2(X, \mathbb{Z}) \longrightarrow H^2(\mathcal{O}_X)$$

$$\downarrow \mathbb{Z}$$

$$H^2(X, \mathbb{C}) \xrightarrow{P} H^{0,2}(X)$$

$$\cong \downarrow \Lambda_2^{-1}$$

$$\cong \downarrow$$

$$\Lambda^2 \Omega \oplus (\Omega \otimes \bar{\Omega}) \oplus \Lambda^2 \bar{\Omega} \longrightarrow \Lambda^2 \bar{\Omega}$$

where

$$\Omega = \text{Hom}_{\mathbb{C}}(\mathbb{C}^n, \mathbb{C})$$

$$\bar{\Omega} = \text{Hom}_{\bar{\mathbb{C}}}(\mathbb{C}^n, \mathbb{C})$$

$$\text{Let } \Lambda_2^{-1} C_1(L) = E = E_1 + E_2 + E_3$$

Since E is $\mathbb{R}$ -valued, $E_3 = \bar{E}_1$.

Since $E \in \ker(H^2(X, \mathbb{Z}) \rightarrow H^2(\mathcal{O}_X))$, $E_3 = 0$.

Then $E = E_2$, i.e. E is $(1,1)$ -form.

*

Such skew form is imaginary part of some Hermitian form.

$$\left[\begin{array}{l} H: \text{hermitian form} \\ H(\Delta, \Delta) \leq \mathbb{Z} \end{array} \right] \xleftrightarrow{1:1} \left[\begin{array}{l} E: \text{real valued skew sym form} \\ E(\Delta, \Delta) \leq \mathbb{Z} \text{ \& } E(iu, iw) = E(u, w) \end{array} \right]$$

$$H(u, w) = E(iu, w) + iE(u, w)$$

$$E(u, w) = \text{Im } H(u, w).$$

Then we can identify $\left[\begin{array}{c} \boxed{} \xleftrightarrow{\text{above}} \boxed{} \\ \nwarrow \nearrow \\ \boxed{\text{Image of } C_1} \end{array} \right] = NS(X) : \text{Neron Severi gp.}$

§1-3 Semicharacter for H .

Q. How to get line bundle from $\begin{cases} H \\ C_1(L) \\ E \end{cases}$?
 factor of automorphy

Def semicharacter for H .

$$\begin{cases} \chi: \Lambda \rightarrow U(1) = \{z \in \mathbb{C} \mid |z| = 1\} \\ \text{s.t. } \chi(\lambda + \mu) = \chi(\lambda) \chi(\mu) \exp(\pi i \operatorname{Im} H(\lambda, \mu)) \end{cases}$$

Let $P(\Lambda) = \{(H, \chi) \mid H \in NS(X), \chi = \text{semichar. for } H\}$.

$$w/ (H_1, \chi_1) \circ (H_2, \chi_2) = (H_1 + H_2, \chi_1 \chi_2)$$

($\sim$ this corresponds to tensor product of line bundles.)

Then we have

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}(\Lambda, U(1)) & \longrightarrow & P(\Lambda) & \xrightarrow{p} & NS(X) \longrightarrow 0 \\ & & \chi & \longmapsto & (0, \chi) & & \star \\ & & & & (H, \chi) & \longmapsto & H \end{array}$$

$\star$ surjectivity of p .

we define for $(H, \chi) \in P(\Lambda)$

$$\alpha_{(H, \chi)}(\lambda, \nu) := \chi(\lambda) \exp(\pi H(\nu, \lambda) + \frac{\pi}{2}(\lambda, \lambda))$$

$\alpha_{(H, \chi)}$ is actually a factor of automorphy (check 1-cocycle rel)

Then we get

$$\alpha_{(H, \chi)} \longmapsto L(H, \chi) := \mathbb{C}^n \times \mathbb{C} / \Lambda$$

$$w/ \lambda \circ (\nu, t) = (\lambda + \nu, \alpha_{(H, \chi)}(\lambda, \nu) \cdot t)$$

Lem the map $P(\Lambda) \longrightarrow \text{Pic}(X)$ is homomorphism and

$$\begin{array}{ccc} & \mathbb{C} & \\ p \swarrow & & \searrow c_1 \\ & NS(X) & \end{array} \text{ is comm.}$$

pf [claim] $c_1(L(H, \chi)) = H$ for all $(H, \chi) \in P(\Lambda)$

$$\text{we write } \chi(\lambda) = \exp(2\pi i \varphi(\lambda))$$

6

we have $\alpha_{(H, \chi)}(\lambda, \mu) = \exp(2\pi i \underline{g}(\lambda, \mu))$

where $g(\lambda, \mu) = \varphi(\lambda) - \frac{i}{2} H(\mu, \lambda) - \frac{i}{4} H(\lambda, \lambda)$

Then we obtain

$$\begin{aligned} F_L(\lambda, \mu) &= g(\mu, \mu + \lambda) + g(\lambda, \mu) - g(\lambda, \mu + \mu) - g(\mu, \mu) \\ &= \frac{i}{2} (H(\lambda, \mu) - H(\mu, \lambda)) = \text{Im } H(\lambda, \mu) \quad \# \end{aligned}$$

Here we use fact

$$\boxed{\text{Hom}(\Lambda, \mathcal{O}(1)) \xrightarrow{\sim} \text{Pic}^0(X) := \ker(C_1)}$$

see [BL] Prop 2.2.2 or [M] p.20.

Then we have Apell-Humbert thm.

Thm

$$\begin{array}{ccccccc} 0 \rightarrow \text{Hom}(\Lambda, \mathcal{O}(1)) & \rightarrow & P(\Lambda) & \rightarrow & NS(X) & \rightarrow & 0 \\ & \searrow \downarrow & & \searrow \downarrow & & \parallel & \\ 0 \rightarrow \text{Pic}^0(X) & \rightarrow & \text{Pic}(X) & \rightarrow & NS(X) & \rightarrow & 0 \end{array}$$

Application of this, see [BL] §2.3 or §2.5.

§2 Cohomology of line bundle

① $H^1(L) = \text{Im } H$ has central information of L .

Frobenius lemma (see Olivier Debarre "Complex tori and Abelian var." Prop 6.1)

$E: \Lambda \times \Lambda \rightarrow \mathbb{Z}$ skew sym.

$\exists$ basis $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ of Λ w.r.t which E is given by

$$\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix} \quad \text{where } D = \text{diag}(d_1, \dots, d_g)$$

$d_i \geq 1$ s.t. $d_i \mid d_{i+1}$

- Rmk
- lemma suggest $\det E \geq 0$.
 - $\lambda_1, \dots, \lambda_g$ is basis of $\mathbb{C}^n$ over $\mathbb{C}$.

$$\leadsto \Lambda = \mathbb{Z}^g \oplus \mathbb{Z} \mathbb{Z}^g$$

where τ : period matrix

w/ τ : symm & $\text{Im}(\tau)$: positive def.

Analogue $g=1$

$$\left\{ \begin{array}{l} \mathbb{C}/\mathbb{Z} + \tau\mathbb{Z} \text{ w/} \\ \text{Im} \tau > 0 \end{array} \right\}$$

We can classify L w.r.t type of H .

$$\left\{ \begin{array}{l} H=0 \end{array} \right\} \begin{cases} \text{trivial} \\ \text{Pic}^0(X) \end{cases}$$

$$\left\{ \begin{array}{l} H \neq 0 \end{array} \right\} \begin{cases} \text{degenerate} \\ \text{non-degenerate} \end{cases}$$

$\left\{ \begin{array}{l} \text{positive def} \\ \text{negative def} \\ \text{otherwise} \end{array} \right\}$

Thm (Lefschetz)

$C_{\text{px tori}}$ is proj

$$\Leftrightarrow \exists L: \text{positive def.}$$

this can be checked
by Riem. Relation

[BL] §4.2.

Trivial

$$H^i(X, \mathcal{O}_X) = \bigwedge^i \text{Hom}_{\mathbb{C}}(\mathbb{C}^n, \mathbb{C})$$

see [M] p.4

$\text{Pic}^0(X) \setminus \text{trivial}$

$$H^i(X, L) = 0 \quad \forall i \quad \text{i.e. topologically triv}$$

see [M] p.72.

Non-degenerate

$i(L) := \# \text{ negative eigen value of } H$: index of L

$$h^p(X, L) = \begin{cases} 0 & p \neq i \\ \sqrt{\det E} = d_1 \cdots d_g & p = i \end{cases}$$

plattian of E

degenerate (Mumford-Kempf vanishing Thm)

$$h^s(L) = \begin{cases} \binom{g-r-s}{g-s} h^s(L) & s \leq g \leq g-r \\ 0 & g < r, g-r > g \end{cases}$$

where $h^s(L) = \begin{cases} \text{Pfr}(E) & \text{if } L \mid k(L)_0 \text{ triv} \\ 0 & \text{otherwise} \end{cases}$ conn comp of $k(L) = \ker \phi_L$

reduced Pfaffian for $(d_1, \dots, d_s, 0, \dots, 0)$

$$\text{Pfr}(E) = \begin{cases} \prod_{v=1}^s d_v & s \neq 0 \\ 1 & s = 0 \end{cases}$$

Example

positive def = ample

what we want to show.

$$h^0(L) = d_1 - dg$$

(idea) $H^0(L) = H^0(\mathcal{O}_{\mathbb{C}^n})^A$

$$= \{ \theta : \text{hol} \mid \theta(u+\lambda) = \chi(\lambda) \exp(\pi H(u, \lambda) + \frac{i}{2}(\lambda, \lambda)) \theta(u) \}$$

$$= \{ \text{theta func's for the factor } a_L \}$$

Then $h^0(L) = \dim \{ \text{space of theta func's} \}$

In particular, for $d_1 = \dots = dg = 1$ (unimodular principally polarized)

$\leadsto$ we get classical Riemann theta function.

$$\theta(z) = \sum_{m \in \mathbb{Z}^g} \exp \left(t_m z + \frac{1}{2} t_m z_m \right)$$

See Arnaud Beauville "classical theta functions and their generalization"